Helix Research Technology LLC

Proposal in response to Solicitation 75N910-27-R-0044 — Research Data Platform and Analytics Support

Department of Health and Human Services — National Institutes of Health

Volume I — Technical

1. Data Platform Architecture

Helix proposes a FedRAMP High research data platform built on a lakehouse architecture with separate zones for raw, curated and analysis-ready data, and with compute that scales to zero between analyses to control cost.

All datasets are registered in a searchable catalog with lineage from source instrument or study to every derived table.

2. Privacy and Controlled-Access Data Protection

Controlled-access human-subjects data is held in a separate enclave. Access follows a dbGaP-aligned workflow: a researcher requests a dataset, the data access committee approves the request, and the platform issues time-bound, purpose-limited access that expires automatically.

No controlled-access data can be exported from the enclave; results leave only after automated disclosure review and, where required, a human privacy officer check.

3. Researcher Self-Service Analytics

Researchers use managed notebooks and a no-code cohort builder against analysis-ready data, with shared, versioned environments so analyses are reproducible.